

Bloom Filter-Based MPSI

Weekly Progress Meeting 7 May 2026

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Bibliography

- [1] Jelle Vos, Jorrit van Assen, Tjitske Koster, Evangelia Anna Markatou, and Zekeriya Erkin. On the insecurity of bloom filter-based private set intersections. *Cryptology ePrint Archive*, 2024.
- [2] Ou Ruan, Changwang Yan, Jing Zhou, and Chaohao Ai. A practical multiparty private set intersection protocol based on bloom filters for unbalanced scenarios. *Applied Sciences*, 13(24):13215, 2023.
- [3] Florian Kerschbaum. Outsourced private set intersection using homomorphic encryption. In *Proceedings of the 7th ACM Symposium on Information, Computer and Communications Security*, pages 85–86, 2012.
- [4] Falzon, Francesca, and Evangelia Anna Markatou. "Re-visiting authorized private set intersection: A new privacy-preserving variant and two protocols." *Proceedings on Privacy Enhancing Technologies* (2025).
- [5] Koster, Tjitske Ollie, Francesca Falzon, and Evangelia Anna Markatou. "Bandwidth Efficient Partial Authorized PSI." *Cryptology ePrint Archive* (2025).
- [6] Lindell, Yehuda. "How to simulate it—a tutorial on the simulation proof technique." *Tutorials on the Foundations of Cryptography: Dedicated to Oded Goldreich* (2017): 277-346.

Progress

- Writing!
- Implement APSI4/PAPSI4/**PAPSI5**

Progress

MAPSI – Multi-Party Authorized Private Set Intersection

Authorization

- Send pk to clients?

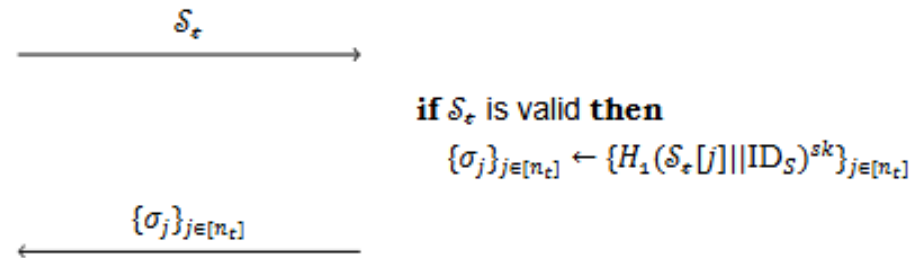
Parameters

Number of parties t ; Party i 's set size $n_i = |\mathcal{S}_i|, \forall i \in [t]$; Groups $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ of prime order q , with generators g_1, g_2, g_T ; Bilinear function $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ such that $e(g_1, g_2) = g_T$; Hash function $H1 : \{0, 1\}^* \rightarrow \mathbb{G}_1$; Judge's public key $pk \leftarrow g_2^{sk}$, where sk is the Judge's private key.

Authorize

Server $\mathcal{S}_\epsilon$

Judge $\mathcal{J}$



Progress

MAPSI

We cannot truncate/hash the pairings - we need the homomorphically multiplicative property

Solution: instead of everyone sending to the leader client (star topology), the clients send from one to the other and finally the last one sends it to the server (ring topology)

- This distributes the work more evenly instead of putting all the burden on the leader
- The highest point in the communication graph will be 100 times lower 😊
- However, computation time will increase, since clients will have to wait for one another ($t - 1$ sequential rounds)

Intersect
Server S_t

Client 1 S_1 ... **Client i** S_i

$S_1 \leftarrow \mathbb{Z}_q$ $S_i \leftarrow \mathbb{Z}_q$
 $S_1 \leftarrow g_2^{S_1}$ $S_i \leftarrow g_2^{S_i}$
 $\text{GBF}_1 \leftarrow \text{InitializeGBF}(m, k)$ $\text{GBF}_i \leftarrow \text{InitializeGBF}(m, k)$
for $j \in [n_1]$ **do** **for** $j \in [n_i]$ **do**
 $\hat{y}_{1,j} \leftarrow e(H_1(S_1[j] || \text{ID}_S), pk^{S_1})$ $\hat{y}_{i,j} \leftarrow e(H_1(S_i[j] || \text{ID}_S), pk^{S_i})$
 $\text{AddToGBF}(\text{GBF}_1, S_1[j], \hat{y}_{1,j})$ $\text{AddToGBF}(\text{GBF}_i, S_i[j], \hat{y}_{i,j})$

$\leftarrow S_2, \text{GBF}_2$

$\leftarrow S_i, \text{GBF}_i$

$S_{\text{agg}} \leftarrow \prod_{i=1}^{t-1} S_i$

$\text{GBF}_{\text{agg}} \leftarrow \text{InitializeGBF}(m, k)$
for $v \in [m]$ **do**
 $\text{GBF}_{\text{agg}}[v] \leftarrow \prod_{i=1}^{t-1} \text{GBF}_i[v]$

$\leftarrow S_{\text{agg}}, \text{GBF}_{\text{agg}}$

$I = \{\}$
for $j \in [n_t]$ **do**
 $\hat{x}_j \leftarrow e(\sigma_j, S_{\text{agg}})$
 $\hat{y}_j \leftarrow \prod_{u=1}^k \text{GBF}_{\text{agg}}[h_u(S_t[j])]$
if $\hat{x}_j = \hat{y}_j$ **then**
 $I = \{S_t[j]\} \cup I$
 Output I

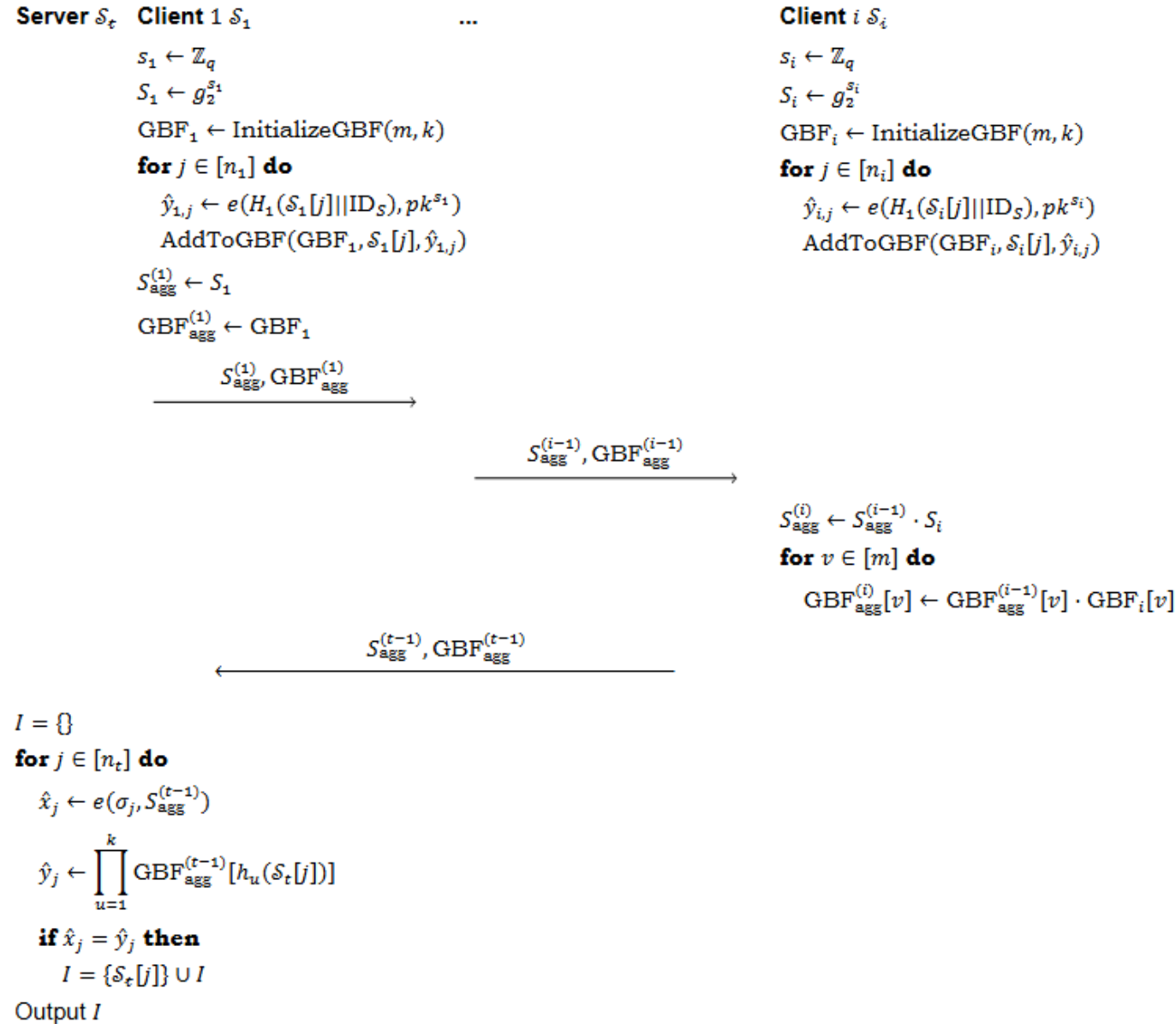
Progress

MAPSI

The clients send from one to the other and finally the last one sends it to the server (ring topology)

- This distributes the work more evenly instead of putting all the burden on the leader
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- However, **computation time will increase**, since clients will have to wait for one another ($t - 1$ sequential rounds)

Intersect



Progress

MAPSI

Proof outline against a semi-honest client leader:

- **H0:** Real protocol
- **H1:** Replace target values $\widehat{y}_{i,j}$ with random elements from $\mathbb{G}_T$
 - The leader will have to distinguish between $e(H_1(\mathcal{S}_i[j]||ID_S), g_2^{sk \times s_i})$ and a random element from $\mathbb{G}_T$, having $S_i = g_2^{s_i}$ and $pk = g_2^{sk}$ - Decisional Bilinear Diffie Hellman (DBDH)
- **H2:** Replace S_i + all bins in the GBFs with random elements from $\mathbb{G}_T$

Intersect
Server S_e

Client 1 S_1

$s_1 \leftarrow \mathbb{Z}_q$
 $S_1 \leftarrow g_2^{s_1}$
 $GBF_1 \leftarrow \text{InitializeGBF}(m, k)$
for $j \in [n_1]$ **do**
 $\hat{y}_{1,j} \leftarrow e(H_1(S_1[j]||ID_S), pk^{s_1})$
 AddToGBF($GBF_1, S_1[j], \hat{y}_{1,j}$)

...

Client i S_i

$s_i \leftarrow \mathbb{Z}_q$
 $S_i \leftarrow g_2^{s_i}$
 $GBF_i \leftarrow \text{InitializeGBF}(m, k)$
for $j \in [n_i]$ **do**
 $\hat{y}_{i,j} \leftarrow e(H_1(S_i[j]||ID_S), pk^{s_i})$
 AddToGBF($GBF_i, S_i[j], \hat{y}_{i,j}$)

S_2, GBF_2

S_i, GBF_i

$S_{\text{agg}} \leftarrow \prod_{i=1}^{t-1} S_i$

for $v \in [m]$ **do**
 $GBF_{\text{agg}}[v] \leftarrow \prod_{i=1}^{t-1} GBF_i[v]$

$S_{\text{agg}}, GBF_{\text{agg}}$

$I = \{\}$
for $j \in [n_r]$ **do**
 $\hat{x}_j \leftarrow e(\sigma_j, S_{\text{agg}})$
 $\hat{y}_j \leftarrow \prod_{u=1}^k GBF_{\text{agg}}[h_u(S_t[j])]$
if $\hat{x}_j = \hat{y}_j$ **do**
 $I = \{S_e[j]\} \cup I$
 Output I

Progress

MPAPSI – Multi-Party Partial Authorized Private Set Intersection

- 2 variants, same authorization

Parameters

Number of parties t ; Party i 's set size $n_i = |\mathcal{S}_i|, \forall i \in [t]$; Groups $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ of prime order q , with generators g_1, g_2, g_T ; Bilinear function $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ such that $e(g_1, g_2) = g_T$; Hash functions $H1 : \{0, 1\}^* \rightarrow \mathbb{G}_1$ and $H2 : \mathbb{G}_1 \rightarrow \mathbb{G}_1$; Judge's public key $pk \leftarrow g_2^{sk}$, where sk is the Judge's private key.

Authorize

Server $\mathcal{S}_t$

Judge $\mathcal{J}$

$r \leftarrow \mathbb{Z}_q$

$\hat{\mathcal{S}}_t \leftarrow \{H_1(\mathcal{S}_t[i] || \text{ID}_S)^r\}_{i \in [n_t]}$

$\hat{\mathcal{S}}_t$

$\mathcal{J}$

$\mathcal{J} \subseteq_S [n]$ st $|\mathcal{J}| = pn$

$\pi \leftarrow \text{EEA.Prove}(r, \{\mathcal{S}_t[i]\}_{i \in \mathcal{J}})$

$\pi, \{\mathcal{S}_t[i]\}_{i \in \mathcal{J}}$

if $\mathcal{S}_t$ is valid then

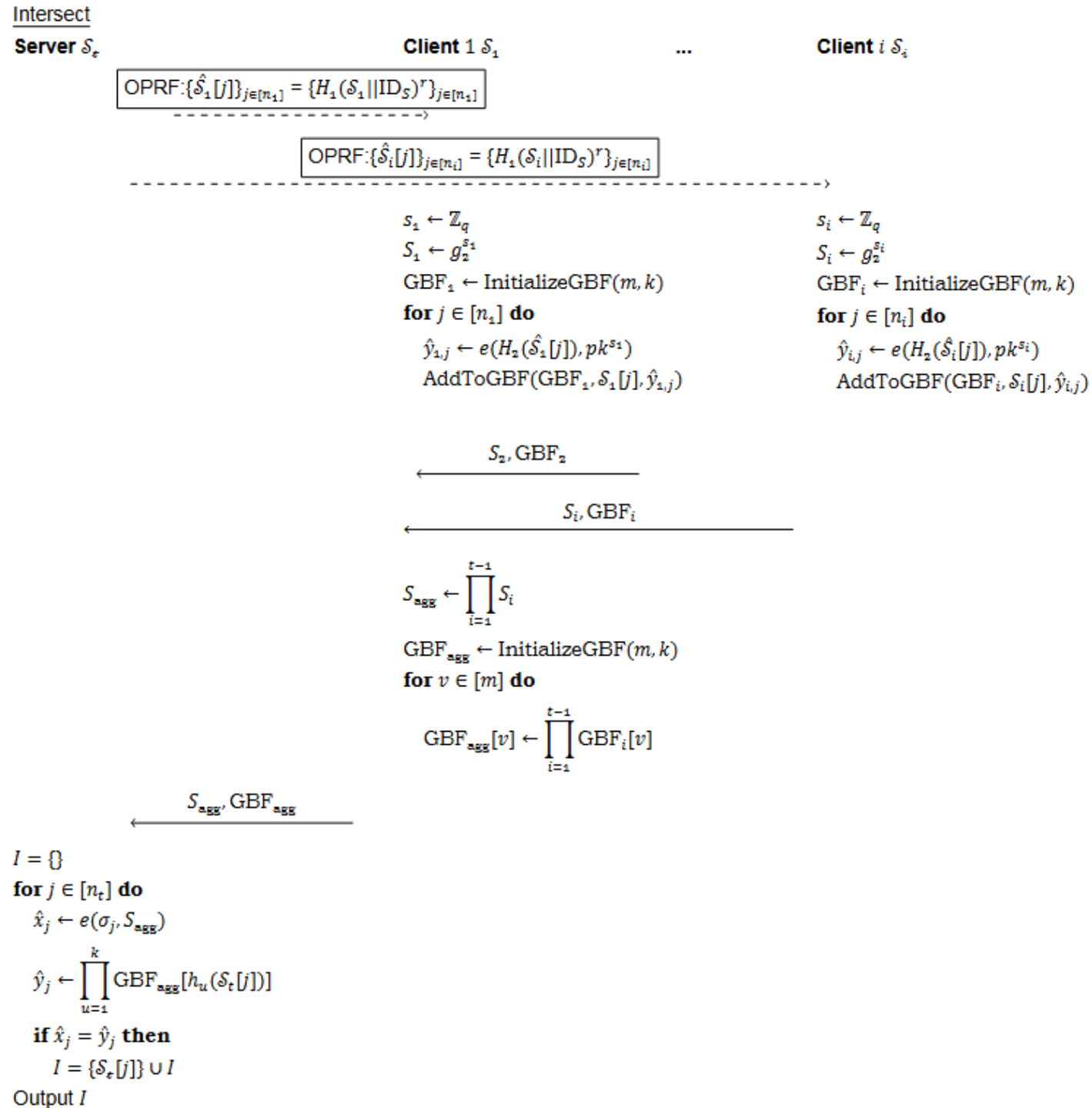
$\{\sigma_j\}_{j \in [n_t]} \leftarrow \{H_2(H_1(\mathcal{S}_t[j] || \text{ID}_S)^r)^{sk}\}_{j \in [n_t]}$

$\{\sigma_j\}_{j \in [n_t]}$

Progress

MPAPSI 1 – Multi-round intersection

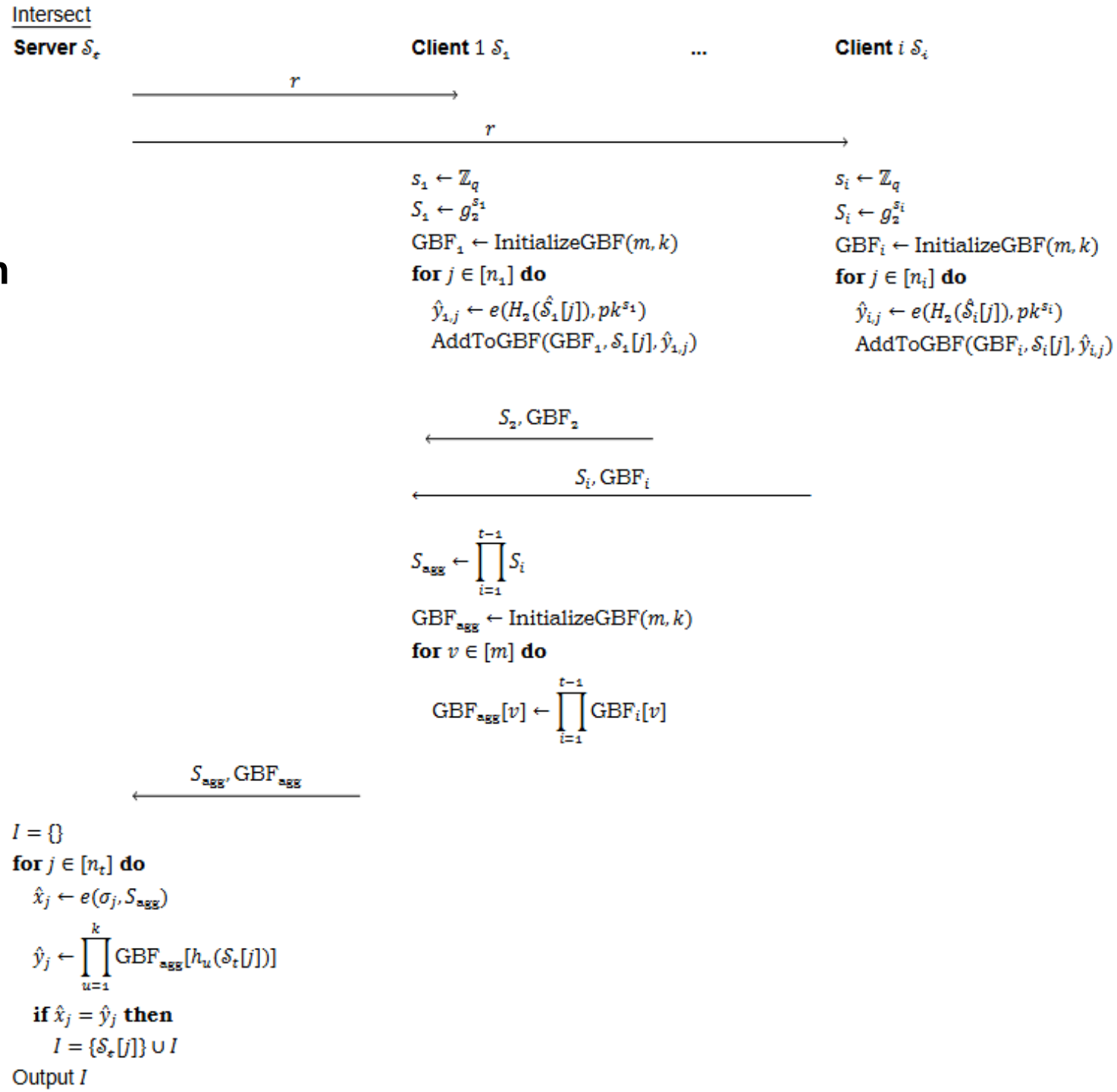
- OPRF with each client



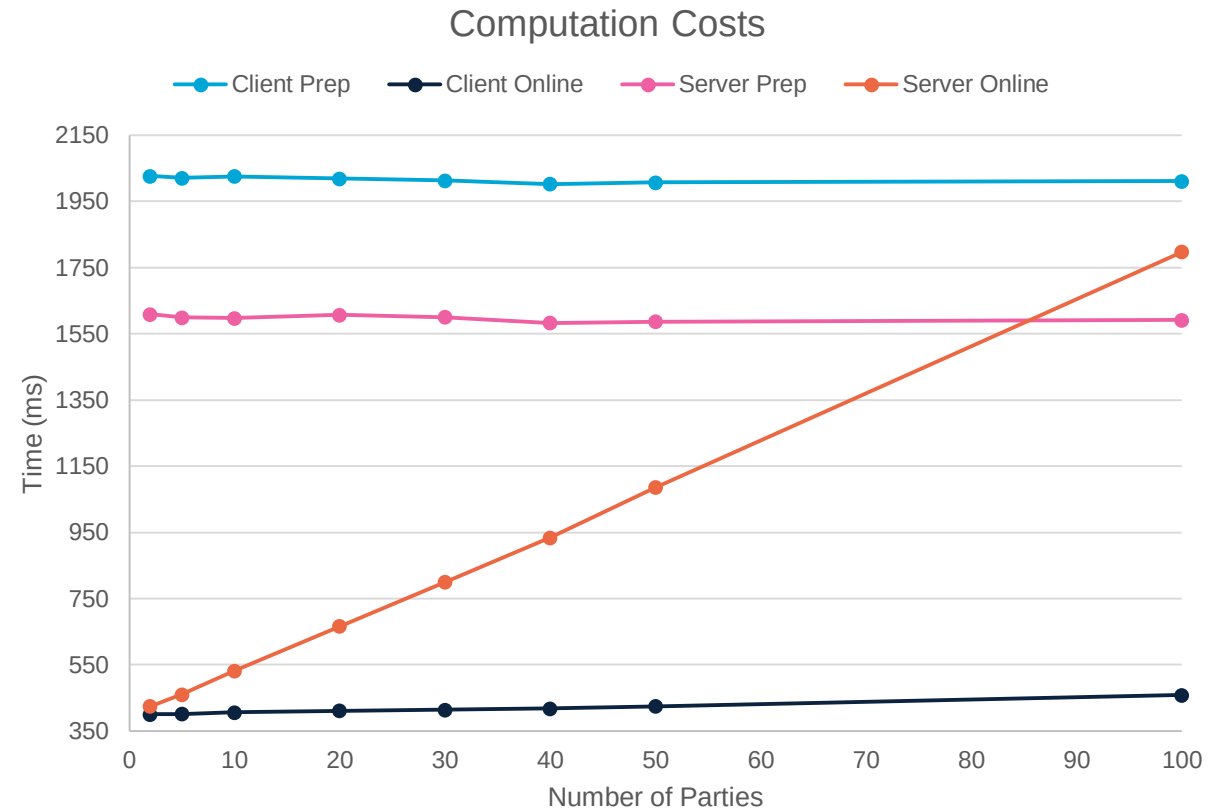
Progress

MPAPSI 2 – Single-round intersection

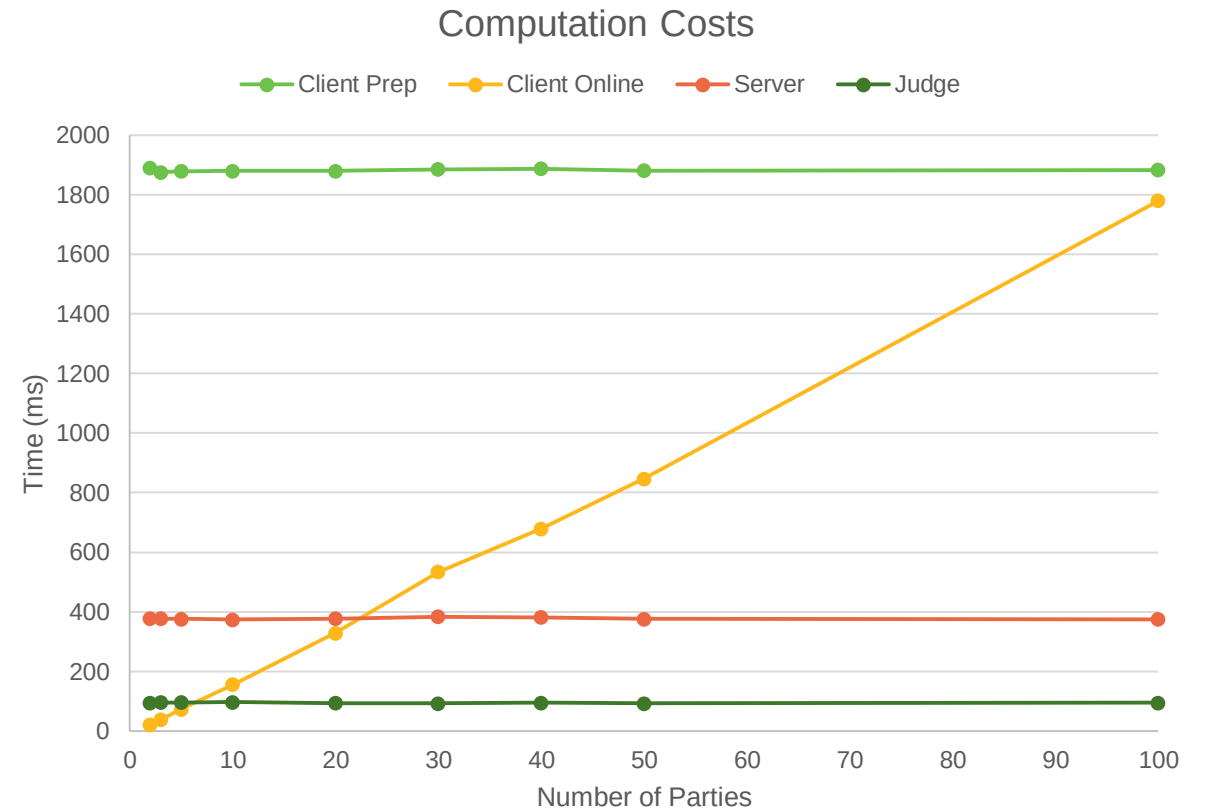
- Sending r directly



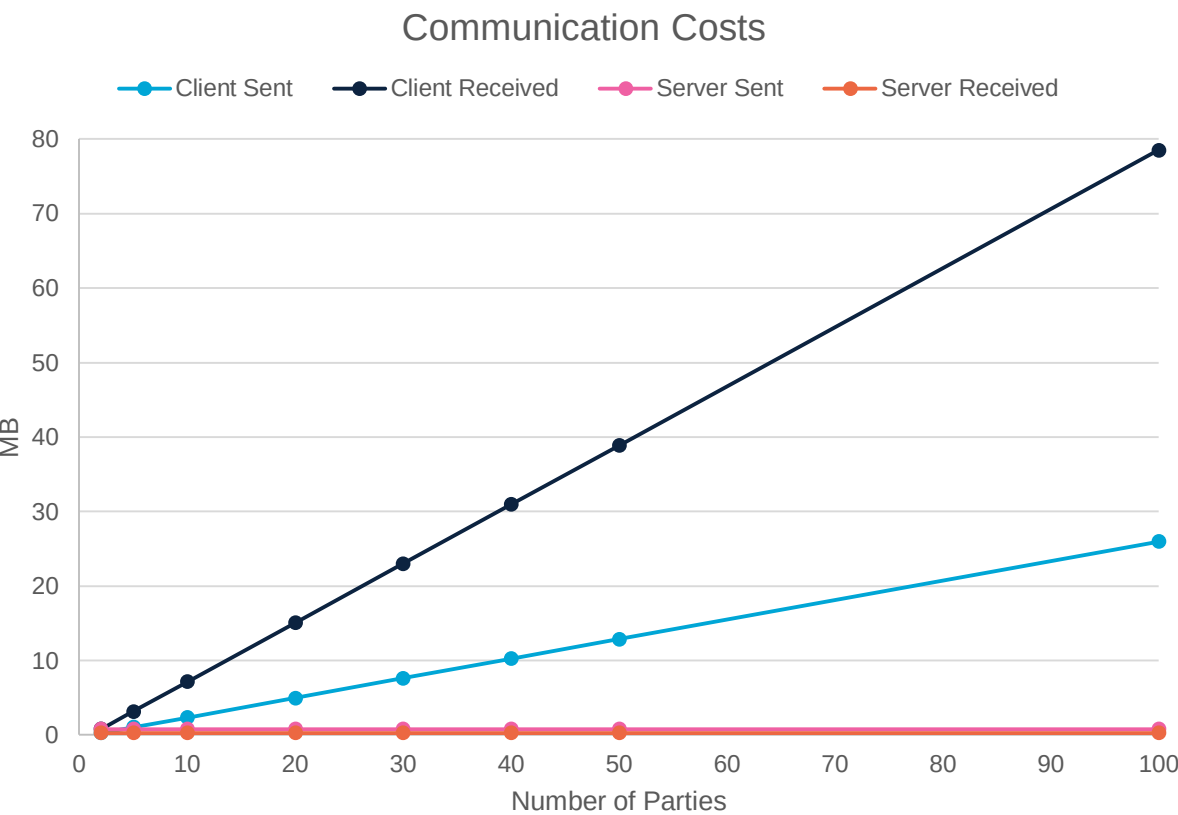
Baseline



MAPSI



Baseline

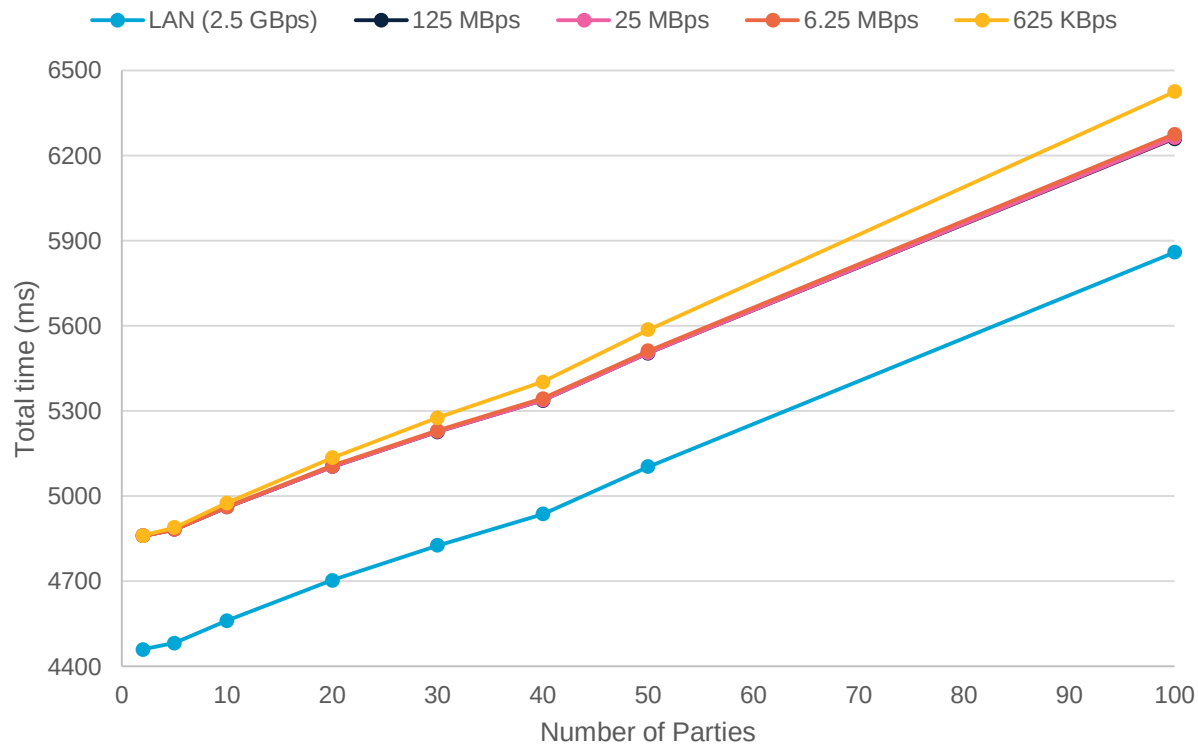


MAPSI



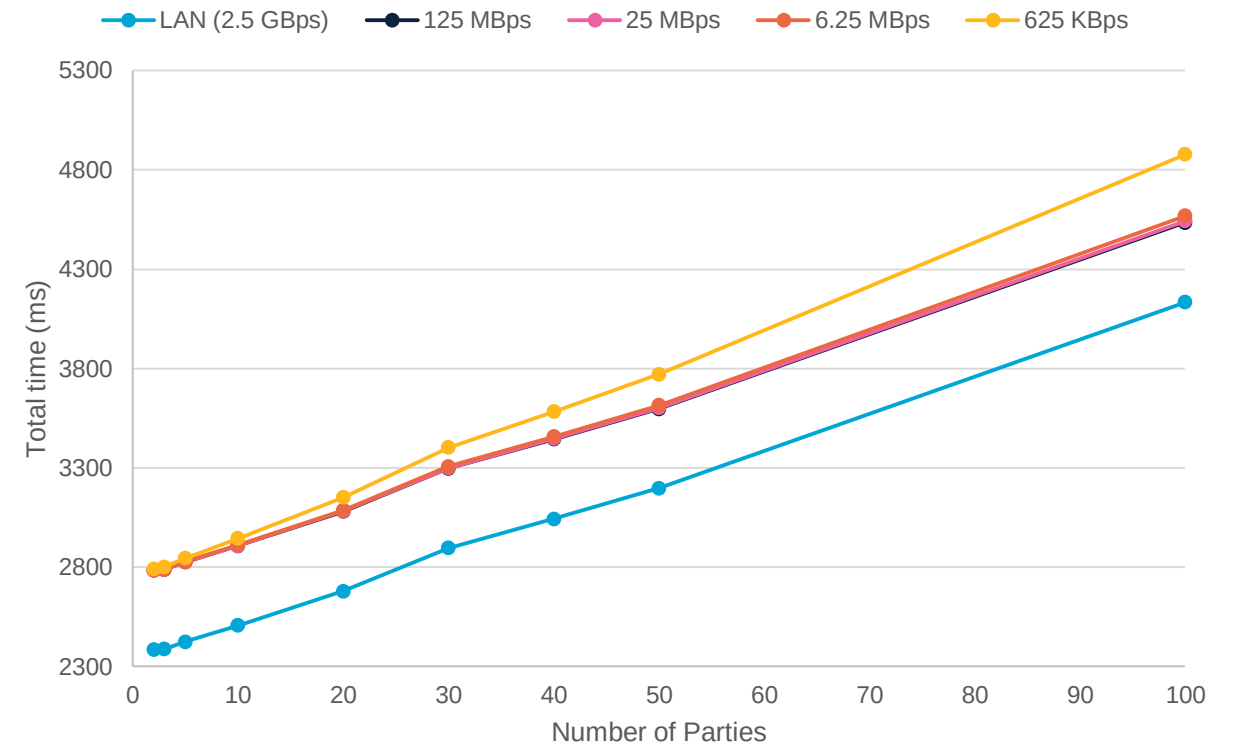
Baseline

Simulated Total Execution Time



MAPSI

Simulated Total Execution Time



Next Week

- Writing!
 - Proofs: security and correctness for MAPSI, MPAPSI1 and MPAPSI2
 - Protocol steps
 - Prerequisites
- Multithreading and benchmarking
 - Comparisons?
 - Leaked password database

Thank you!

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